

Xiaoxuan Wang

✉ xw27@cs.ucla.edu | 🎓 Google Scholar | 🌐 mandyyyyii.github.io

OBJECTIVE

My research centers on **Agentic Reinforcement Learning** and **LLM agents** — building stable, scalable agents for real-world interaction. I am also interested in **LLM reasoning**, spanning planning, tool use, and scientific problem-solving. I organize the UCLA DM reading group.

EDUCATION

University of California, Los Angeles

Sept. 2022 – Present

Ph.D. in Computer Science Advisor: Wei Wang

Los Angeles, CA

University of Illinois Urbana-Champaign

Aug. 2018 – May 2022

B.S. in Computer Science, Grainger College of Engineering

Urbana, IL

- Minor in Mathematics James Scholar Honor Dean's List

SELECTED PROJECTS

ARLarena: A Unified Framework for Stable Agentic Reinforcement Learning

2025 – Present

Project Lead

UCLA

- Leading a large-scale project on agentic LLMs, developing principled training and inference recipes across 8+ tasks.
- Building a unified and extensible pipeline supporting 16+ policy optimization algorithms, with extensions to asynchronous agent training and MoE models.
- Developing a multi-agent system for solving interactive agentic tasks and extending it to support multi-agent training.
- Conducting extensive empirical analyses of training stability, scalability, and system efficiency, including cross-hardware comparisons across NVIDIA and AMD GPUs.

SciBench: Evaluating Scientific Problem-Solving in Large Language Models

2023 – 2024

- Developed SciBench to evaluate college-level scientific reasoning and problem solving in LLMs.
- Designed evaluation protocols and analyses for model capabilities and failure modes across scientific domains.
- Published at *ICML 2024*; received media coverage in a **Nature News Feature**.

PUBLICATIONS

Agentic RL & LLM Agents

- **HarnessBridge: Learnable Bidirectional Controller for LLM Agent Harness**
Xiaoxuan Wang*, Haixin Wang*, Alex Taylor, Jason Cong, Yizhou Sun, Wei Wang
- **ARLarena: A Unified Framework for Stable Agentic Reinforcement Learning**
Xiaoxuan Wang*, Han Zhang*, Haixin Wang*, Yidan Shi[†], Ruoyan Li[†], Kaiqiao Han[†], Chengyi Tong, Haoran Deng, Alex Taylor, Yanqiao Zhu, Renliang Sun, Jason Cong, Yizhou Sun, Wei Wang
ICML 2026

LLM Reasoning, Post-Training & Evaluation

- **SciBench: Evaluating College-Level Scientific Problem-Solving Abilities of Large Language Models**
Xiaoxuan Wang*, Ziniu Hu*, Pan Lu*, Yanqiao Zhu*, Jieyu Zhang, Satyen Subramaniam, Arjun R. Loomba, Shichang Zhang, Yizhou Sun, Wei Wang
ICML 2024 Media Coverage: **Nature News Feature**
- **EAST: Entropy-Based Adaptive Weighting for Self-Training**
Xiaoxuan Wang, Yihe Deng, Mingyu Derek Ma, Wei Wang
NEURIPS MATHAI, 2025
- **From Solving to Verifying: A Unified Objective for Robust Reasoning in LLMs**
Xiaoxuan Wang, Bo Liu, Song Jiang, Jingzhou Liu, Jingyuan Qi, Xia Chen, Baosheng He
- **MatSciBench: Benchmarking the Reasoning Ability of Large Language Models in Materials Science**
Junkai Zhang*, Jingru Gan*, Xiaoxuan Wang, Zian Jia, Changquan Gu, Jianpeng Chen, Yanqiao Zhu, Mingyu Derek Ma, Dawei Zhou, Ling Li, Wei Wang
KDD 2026

Clinical & Domain Applications

- **CliBench: Multifaceted Evaluation of LLMs in Clinical Decision Making**
Mingyu Derek Ma, Chenchen Ye, Yu Yan, Xiaoxuan Wang, Peipei Ping, Timothy S. Chang, Wei Wang
- **Memorize and Rank: Evaluating LLMs for Clinical Diagnosis Prediction**

Mingyu Derek Ma, **Xiaoxuan Wang**, Yijia Xiao, Anthony Cuturrufo, Wei Wang
AAAI 2025; NeurIPS GenAI4Health 2024

Earlier Work (NLP & Speech)

- **STAR: Boosting Low-Resource Event Extraction via Structure-to-Text Generation**
Mingyu Derek Ma, **Xiaoxuan Wang**, Po-Nien Kung, P. Jeffrey Brantingham, Nanyun Peng, Wei Wang
AAAI 2024
- **Learning under Label Proportions for Text Classification**
Jatin Chauhan, **Xiaoxuan Wang**, Wei Wang
EMNLP Findings 2023
- **Global Responses to the COVID-19 Pandemic: Evidence Finding and Verification**
Rotem Dror, **Xiaoxuan Wang**, Dan Roth
- **Seamless Equal Accuracy Ratio for Inclusive CTC Speech Recognition**
Heting Gao, **Xiaoxuan Wang**, Sunghun Kang, et al.
Speech Communication, 2022

INVITED TALKS

ARLArena: A Unified Framework for Stable Agentic Reinforcement Learning, NICE (Nexus for IntelligenCE), March 2026

INTERNSHIP EXPERIENCE

Meta Platforms, Inc.

Summer 2025

Research Intern (AI)

- Developed **GRPO-Verif**, a novel reinforcement learning algorithm that jointly optimizes solution generation and self-verification in LLMs via a unified loss formulation.
- Demonstrated improved self-verification while preserving reasoning performance, with tunable hyperparameters controlling verification signal strength.
- Conducted large-scale empirical evaluations validating robustness and generalization.

Amazon Development Center U.S., Inc.

Summer 2024

Applied Scientist Intern

- Implemented and evaluated RLHF methods including REST-EM, Iterative DPO, and variants for tool-integrated problem solving in LLMs.
- Identified sensitivity of REST-EM to data quality and quantity relative to online preference optimization methods.
- Showed that regularization mitigates overfitting from self-generated data and improves tool usage through controlled log-probability optimization.

SKILLS

Research Areas: Agentic RL, Coding Agents & Agent Harness Design, Post-Training, LLM Reasoning, RLHF

Languages: Python, C++, Java, SQL

Frameworks: PyTorch, Ray, vLLM/SGLang, Megatron-LM, VERL, HuggingFace, FSDP